

Engineering Geology

— Unit 1 Notes

1. Scope of Geology (in Civil Engineering)

Definition: Geology is the science that deals with the study of the Earth — its origin, composition, structure, processes, and the history of life on it.

Engineering Geology is the branch that applies geological knowledge and principles to civil engineering practice.

Scope covers:

- Study of rocks and soils as **construction materials** (strength,

durability, workability).

- Study of rocks and soils as **foundation material** for dams, bridges, buildings, and towers.
- Investigation of **groundwater** — occurrence, movement, and its effect on structures and excavation (dewatering).
- Study of **geological structures** (folds, faults, joints, unconformities) that affect stability of structures, tunnels, and slopes.
- Assessment of **natural hazards** — landslides, earthquakes, subsidence, erosion — for site selection and hazard mitigation.
- Selection of sites for **dams, reservoirs, tunnels, highways, bridges, and railways** based on rock/soil suitability.

- Study of **weathering** processes and their effect on rock mass strength.
- **Quarrying** and selection of sites for building stone, aggregates, and dimension stone.
- **Seismic zoning** and earthquake-resistant design inputs.
- **Coastal and river engineering** — erosion, sedimentation, and flood-plain behavior.

Why it matters: Almost every civil engineering structure ultimately rests on, or is built from, geological material. A structure is only as good as the ground beneath it — hence "the foundation of civil engineering is geology."

2. Subdivisions (Branches) of Geology

Geology is a vast science and is divided into several specialized branches, many of which are directly useful to civil engineers:

Branch	Deals With
Physical Geology	Earth's physical features, processes (weathering, erosion, deposition) that shape landforms
Mineralogy	Study of minerals — composition, physical & chemical properties, classification
Petrology	Study of rocks — origin, mode of

Branch	Deals With
	formation, composition, classification
Structural Geology	Study of deformation of rocks — folds, faults, joints, unconformities
Stratigraphy	Study of layered (sedimentary) rocks, their sequence, age, and correlation
Palaeontology	Study of fossils and ancient life forms, used for dating rock strata
Geomorphology	Study of landforms and the

Branch	Deals With
	processes that create them
Economic Geology	Study of ore deposits, minerals, and fuels of economic value
Hydrogeology	Study of groundwater — occurrence, movement, quality
Seismology	Study of earthquakes, seismic waves, and Earth's interior via wave behavior
Engineering Geology	Application of geology to engineering problems

Branch	Deals With
	(foundations, tunnels, dams, slopes)
Mining Geology	Study related to exploration and extraction of minerals
Photogeology / Remote Sensing	Study of Earth's surface using aerial photographs and satellite imagery

Civil engineering relevance: Structural geology (stability of rock slopes/tunnels), hydrogeology (seepage, dewatering, water supply), petrology & mineralogy (construction material selection), and geomorphology

(site evaluation, erosion/flood behavior)
are the most directly applied branches.

3. Interior of the Earth

The Earth is a differentiated body — its interior is layered based on **composition** and **physical state**, known largely through **seismic wave studies** (P and S waves), gravity data, and meteorite studies.

A. Compositional Layers

1. Crust

- Outermost solid layer; thickness varies: ~5–10 km (oceanic crust) to ~30–70 km (continental crust, thicker under mountains).
- **Oceanic crust:** thinner, denser, basaltic (rich in Silica + Magnesium

→ **SIMA**).

- **Continental crust:** thicker, lighter, granitic (rich in Silica + Aluminium → **SIAL**).
- Separated from mantle by the **Mohorovičić Discontinuity (Moho)**.

2. Mantle

- Extends from Moho to ~2,900 km depth.
- Composed mainly of **silicate minerals rich in Iron and Magnesium** (peridotite-like, "SIMA" continued).
- Subdivided into:
 - **Upper Mantle** (up to ~660 km) — includes the **Asthenosphere** (partially molten, plastic/ductile zone on which tectonic plates move).

- **Lower Mantle / Mesosphere**
(660–2900 km) — more rigid despite high temperature, due to high pressure.
- Boundary with core: **Gutenberg Discontinuity**.

3. Core

- Extends from ~2,900 km to the Earth's center (~6,371 km).
- Composed mainly of **Iron and Nickel (NIFE)**.
- **Outer Core** (2900–5150 km): liquid/molten state; responsible for generating Earth's **magnetic field** (geodynamo).
- **Inner Core** (5150–6371 km): solid, despite extremely high temperature, due to immense pressure.

- Boundary between outer and inner core: **Lehmann Discontinuity**.

B. Mechanical (Physical State) Layers

Layer	Depth	Nature
Lithosphere	0–100 km (crust + uppermost rigid mantle)	Rigid, broken into tectonic plates
Asthenosphere	~100–350 km	Plastic, partially molten, plates move over it
Mesosphere	350–2900 km	Rigid lower mantle

Layer	Depth	Nature
Outer Core	2900– 5150 km	Liquid
Inner Core	5150–6371 km	Solid

C. Evidence for Earth's Interior

- **Seismic waves:** P-waves (primary, travel through solids & liquids) and S-waves (secondary, travel only through solids) — S-waves disappearing in the outer core proves it is liquid.
- **Gravity and density studies**
(average Earth density $\sim 5.5 \text{ g/cm}^3$, much higher than surface rocks $\sim 2.7 \text{ g/cm}^3$, indicating a dense core).
- **Meteorite composition** (assumed similar to Earth's core material).

- **Magnetic field studies.**
- **Volcanic/magma studies** for upper mantle characteristics.

D. Temperature and Pressure

- **Geothermal gradient:** temperature increases with depth, roughly 20–30°C per km near the surface (rate decreases with depth).
 - Core temperature estimated at **5000–6000°C**, close to the temperature of the Sun's surface.
 - Pressure increases enormously with depth (up to ~3.6 million atmospheres at the center), which keeps the inner core solid despite the very high temperature.
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4. Importance of Geological Studies in Civil Engineering Projects

Geological investigation is essential at the **planning, design, and construction stages** of nearly every civil engineering project.

A. Dams and Reservoirs

- Foundation rock must be strong, non-porous, and free from major faults/fractures to avoid **seepage and failure**.
- Reservoir site must be checked for **water tightness** (no cavernous limestone, permeable strata) to prevent leakage.
- Study of faults/joints to assess **reservoir-induced seismicity**.

- Assessment of **landslide-prone slopes** around the reservoir rim (e.g., Vaiont Dam disaster, Italy, was caused by a massive landslide into the reservoir).

B. Tunnels

- Rock type, its strength, and structural features (joints, faults, folds) determine tunneling method, support system, and cost.
- Groundwater conditions affect excavation (need for dewatering/grouting).
- Overbreak, rock bursts, and squeezing ground conditions depend on geological structure.

C. Bridges

- Foundation conditions of piers and abutments (bearing capacity of

rock/soil at riverbed).

- Scour potential related to rock/soil erodibility.
- Study of river geomorphology (meandering, aggradation/degradation).

D. Highways and Railways

- Alignment selection avoiding landslide zones, expansive soils, and unstable slopes.
- Selection of quarry sites for road-building material (aggregate, ballast).
- Study of cut-slope stability in hilly terrain.

E. Buildings and Foundations

- Bearing capacity of underlying rock/soil.

- Settlement behavior, especially over weak or compressible strata.
- Effect of groundwater table on foundation design.

F. Reservoirs, Canals, and Irrigation Projects

- Seepage losses through canal beds.
- Suitability of soil/rock for lining.

G. Natural Hazard Mitigation

- Landslide hazard zonation for hill-area construction.
- Earthquake microzonation for seismic design.
- Land subsidence studies (especially in mining areas).

H. Quarrying and Material Investigation

- Identification and testing of rock as source of building stone, road metal, and aggregate.
- Location of sand, gravel, and clay deposits.

In summary: Geological studies help engineers choose the **right site, right material, and right design**, preventing failures, cost overruns, and disasters — making geology a foundational input to safe and economical civil engineering.

5. Departments Dealing with Geology in India and Their Scope of Work

A. Geological Survey of India (GSI)

- **Established:** 1851; one of the oldest geoscientific organizations in the world.
- **Administrative Ministry:** Ministry of Mines, Government of India.
- **Headquarters:** Kolkata, with regional offices across India.

Scope of Work:

- Systematic **geological mapping** of the entire country (regional and detailed surveys).
- **Mineral exploration** and assessment of mineral resources (metallic, non-metallic, fuel minerals).
- **Geotechnical studies** for major engineering projects (dams, tunnels, power projects).
- **Landslide studies** and hazard zonation.

- **Seismotectonic studies** and earthquake-related investigations.
- **Geochemical and geophysical surveys.**
- **Airborne and satellite-based remote sensing surveys.**
- Publishing **geological maps, memoirs, and reports** used by planners and engineers.
- Advisory role to government and public sector on geological matters, including for major infrastructure projects.
- **Training** geologists and maintaining a national repository of geoscience data (National Geoscience Data Repository).

B. Granite Dimension Stone Cell

- Set up under GSI / Ministry of Mines to give special focus to India's **dimension stone (granite, marble) industry** — a major mineral-based export sector.

Scope of Work:

- Exploration and evaluation of **granite and other dimension stone deposits** (color, texture, block-ability, durability).
- Assessment of **quality parameters** — mineral composition, grain size, presence of joints/fractures affecting block size and yield.
- Assistance to the **dimension stone (granite/marble) industry** for resource assessment.
- Promotion of scientific quarrying methods to improve yield and reduce wastage.

- Support to export-oriented granite industry through resource database and technical guidance.

C. National Institute of Rock Mechanics (NIRM)

- Located at **Kolar Gold Fields (KGF), Karnataka.**
- An autonomous R&D institute originally set up to address rock mechanics problems of deep gold mines, now serving the wider mining and civil engineering sector.

Scope of Work:

- Research in **rock mechanics** — stress analysis, rock mass behavior, stability of underground openings.
- **Rock mass classification** and characterization for tunnels, mines, and underground structures.

- Studies on **rock bursts, ground control, and support design** in deep mines/tunnels.
- **Slope stability analysis** for open-pit mines and hill-area civil projects.
- **Blast vibration studies** and controlled blasting techniques.
- Instrumentation and monitoring of underground/surface excavations.
- Consultancy services to mining and civil engineering (tunneling, hydro-power) projects across India.
- Training programs in rock mechanics and geotechnical engineering.

*(Other related bodies worth knowing:
Central Ground Water Board (CGWB)
— groundwater studies; Central Building
Research Institute (CBRI), Roorkee —
building material and foundation
research; Indian Bureau of Mines (IBM)*

— *mineral conservation and mining regulation.*)

6. Use of Geological Maps and Interpretation of Data

A. Geological Maps — Definition

A **geological map** is a graphical representation, on a topographic base, showing the **distribution of rock types, their ages, boundaries, and structural features** (dip, strike, faults, folds) present at or near the Earth's surface.

B. Types of Geological Maps

- **Areal geological maps:** show distribution of rock units over an area.

- **Structural maps:** emphasize folds, faults, joints.
- **Geomorphological maps:** show landforms.
- **Engineering geological maps:** prepared specifically for engineering projects, showing rock quality, foundation conditions, hazard zones.
- **Hydrogeological maps:** show groundwater-related features.

C. Conventional Symbols and Information Shown

- **Rock type/formation** shown by distinct colors and patterns as per a legend (stratigraphic column).
- **Strike and dip** symbols (a long line for strike direction with a short tick and number for dip amount/direction).

- **Faults** shown as bold/dashed lines, often with arrows indicating relative movement.
- **Folds** — anticline and syncline axes marked with special symbols.
- **Unconformities** marked distinctly from normal contacts.
- **Contour lines** (topography) as the base map.
- **Cross-sections** usually accompany the map to show subsurface structure along a chosen line.

D. Use of Geological Maps in Civil Engineering

1. **Site Selection:** Helps identify suitable rock/soil types for dam, bridge, or building foundations and avoid geologically weak zones (faults, loose soil, cavities).

2. **Route Alignment:** For highways, railways, tunnels, and canals — maps help select routes avoiding landslide-prone or highly fractured terrain.
3. **Reservoir Planning:** Identifies water-tight strata versus permeable/porous rock to prevent seepage losses.
4. **Tunnel Alignment:** Predicts rock types to be encountered, likely support requirements, and groundwater problems along the tunnel path.
5. **Quarry Site Identification:** Locates rock outcrops suitable for construction material.
6. **Hazard Assessment:** Identifies fault zones (seismic risk), landslide-prone slopes, and subsidence-prone (e.g., old mining) areas.

7. **Groundwater Exploration:** Identifies aquifers and recharge zones for water supply projects.
8. **Preliminary Cost Estimation:** Rock type and structure influence excavation method (blasting vs. ripping vs. digging), which affects project cost.

E. Interpretation of Geological Map Data

1. Reading Strike and Dip

- **Strike:** direction of the line formed by the intersection of an inclined rock bed with a horizontal plane (compass direction).
- **Dip:** angle of maximum inclination of the bed measured from horizontal, in a direction perpendicular to strike.

- On outcrop patterns, the "**V-rule**": beds forming a "V" pointing upstream when crossing a valley indicate the bed dips in the same direction as the "V" points (dips upstream) if dip is gentler than the stream gradient in certain cases — this rule is used to infer the direction and approximate steepness of dip from outcrop patterns on topography.

2. Identifying Structures from Outcrop Patterns

- **Horizontal beds:** outcrop pattern parallel to contour lines (follows topography).
- **Dipping beds:** outcrop crosses contours in a "V" pattern.
- **Folds:** repetition of the same rock units symmetrically about an axis; older rocks in the core of an

anticline, younger rocks in the core of a **syncline** (in normal, unfolded sequences).

- **Faults:** sudden discontinuity/offset of rock units or abrupt termination of an outcrop belt.
- **Unconformities:** older, often more deformed, rocks are overlain by younger, less deformed beds with a gap in the geological time record; recognized by an angular discordance or a distinct erosional boundary in the map/cross-section.

3. Constructing Cross-Sections

- Using strike-dip data and topographic profile along a chosen line, subsurface structure is projected to show layer geometry, useful for **estimating depth to bedrock, foundation rock type at a given**

depth, and structural features that may be intersected by a tunnel or foundation excavation.

4. Practical Interpretation for Engineering

- **Steeply dipping/highly jointed rocks:** more prone to instability in cuttings and tunnels.
- **Presence of faults near a project site:** requires special design consideration (grouting, avoidance, or reinforced design).
- **Unconformities:** may represent zones of weakness or differing permeability — relevant for seepage studies.
- **Rock type distribution:** guides choice of foundation type (shallow vs. deep/pile foundations) and excavation method.

F. Sources of Geological Maps in India

- Published by the **Geological Survey of India (GSI)** at various scales (e.g., 1:50,000, 1:250,000).
- Supplemented by **site-specific engineering geological maps** prepared during project investigation (often at larger scales, e.g., 1:1000 or 1:2000, for detailed project sites).

End of notes.